SUN DEVIL
ROCKETRY

Avionics Team

New Liquids Feed System (NLFS)
Test Plan

Project	New Liquids Architecture
Version	1.0
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1. SOFTWARE

1.1. Structural Requirement Analysis

Author:	Eli Sells
Executed By:	
Test Date:	
Result:	TODO
Requirements:	RQ.NLFS.00001, RQ.NLFS.00002, RQ.NLFS.00003
Justification/Plan for Remediation (if failed):	

Step 1:

Description:	Verify the programming languages used.
Steps:	Open the LQD-DAQ source code (https://github.com/SunDevilRocketry/LQD-DAQ). Verify that the project is written in Python. Open the Flight-Software-GUI source code (https://github.com/SunDevilRocketry/Flight-Software-GUI). Verify that the project is written in TypeScript.
Requirements:	RQ.NLFS.00001, RQ.NLFS.00002
Result:	
Screenshot(s):	
Justification:	

Step 2:

Description:	Verify that the software's primary interface is a GUI.

Steps:	Start the Liquids Dashboard. Verify that the project opens a GUI pane.
Requirements:	RQ.NLFS.00003
Result:	
Screenshot(s):	
Justification:	

1.2. UI System Test

Author:	Eli Sells
Executed By:	
Test Date:	
Result:	TODO
Requirements:	RQ.NLFS.00004, RQ.NLFS.00005, RQ.NLFS.00006, RQ.NLFS.00014, RQ.NLFS.00042, RQ.NLFS.00043, RQ.NLFS.00044, RQ.NLFS.00035, RQ.NLFS.00036, RQ.NLFS.00037, RQ.NLFS.00038
Justification/Plan for Remediation (if failed):	

Step 1:

Description:	Verify the presence of the P&ID
Steps:	Start the Liquids Dashboard. Verify that the project shows a P&ID.
Requirements:	RQ.NLFS.00004
Result:	

Screenshot(s):	
Justification:	

Step 2:

Description:	Verify adherence to the color coding scheme
Steps:	Verify that: - Sensor readings are displayed in green (under nominal conditions). - Open valves are displayed in green. - Closed valves are displayed in red. Now, open the settings page, and click the “mock” tab. Change the “mock data status” to “caution”. Verify that sensor readings are now displayed in orange/amber. Now, open the settings page, and click the “mock” tab. Change the “mock data status” to “warning”. Verify that sensor readings are now displayed in red.
Requirements:	RQ.NLFS.00005, RQ.NLFS.00006
Result:	
Screenshot(s):	
Justification:	

Step 3:

Description:	Verify presence of the appropriate derived readings.
Steps:	Verify that: - A rolling chart is given for chamber pressure - A rolling chart is given for thrust - A rolling chart is given for LOx pressure
Requirements:	RQ.NLFS.00014, RQ.NLFS.00042, RQ.NLFS.00043,

	RQ.NLFS.00044,
Result:	
Screenshot(s):	
Justification:	

Step 4:

Description:	Verify that the user configurable settings page behaves as expected
Steps:	Open the settings page. Change the system's dark mode setting, aural enablement setting, and all units settings. Attempt to connect to LQD-DAQ at a URL that the service is not running on. Verify that: - The display units have changed on the P&ID, the gauges, and the rolling charts. - The caution ding sounds (if Aurals were enabled) or does not sound (if Aurals were disabled). - The interface's background is white (if dark mode was enabled) or the background is black (if light mode was enabled). Restart the application. Verify that all of the above still apply.
Requirements:	RQ.NLFS.00035, RQ.NLFS.00036, RQ.NLFS.00037, RQ.NLFS.00038, RQ.NLFS.00039
Result:	
Screenshot(s):	
Justification:	

Step 5:

Description:	Verify that the ABORT button highlighting functions as expected.

Steps:	Refresh the page. Verify that: - The abort button is present Open the settings page. Use the “mock” tab to add a warning level system message. Navigate back to the P&ID. Verify that: - The ABORT option is highlighted in orange. Clear the system message. Actuate any valve. Verify that: - The ABORT option is highlighted in yellow. Restart the application. Verify that all of the above still apply.
Requirements:	RQ.NLFS.00008, RQ.NLFS.00009
Result:	
Screenshot(s):	
Justification:	

1.3. UI/LQD-DAQ Integration Test

Author:	Eli Sells
Executed By:	
Test Date:	
Result:	TODO
Requirements:	RQ.NLFS.00012, RQ.NLFS.00013, RQ.NLFS.00029, RQ.NLFS.00040, RQ.NLFS.00049, RQ.NLFS.00050, RQ.NLFS.00056, RQ.NLFS.00057, RQ.NLFS.00058, FC.NLFS.LD.2, FC.NLFS.LQDDAQ.2
Justification/Plan for Remediation (if failed):	

Step 1:

Description:	Verify that connection status is displayed properly.
Steps:	Launch the UI. Launch LQD-DAQ. Verify that: - The UI is shown as disconnected from LQD-DAQ. Connect. Verify that: - The UI is shown as connected to LQD-DAQ. - Debug info is given in the LQD-DAQ log. Kill the LQD-DAQ process (send SIGINT). Verify that: - An alert is triggered - The browser console for the UI shows that connection failed.
Requirements:	RQ.NLFS.00012, RQ.NLFS.00013, RQ.NLFS.00029, RQ.NLFS.00040, FC.NLFS.LD.2
Result:	
Screenshot(s):	
Justification:	

Step 2:

Description:	Verify that control inputs are handled appropriately.
Steps:	Reconnect the UI and LQD-DAQ, and connect the DAQ service to the test cart. Send a control input to a solenoid valve. Verify that: - The color of the valve on the P&ID changes to match the state of the solenoid valve. Send a control input to a main valve. Verify that: - Intermediate feedback is given while the valve moves. Send the abort signal. Verify that: - The display returns all valves to a safe state, and the abort command is processed in LQD-DAQ.
Requirements:	RQ.NLFS.00007, RQ.NLFS.00010, RQ.NLFS.00011,

	RQ.NLFS.00050
Result:	
Screenshot(s):	
Justification:	

Step 3:

Description:	Verify that logging works properly.
Steps:	Restart the test cart, UI, and LQD-DAQ. Begin a sequence and run it through to completion. Verify that: - A data log was generated - The generated data log has a time delta of 5 ms between samples (and is therefore greater than 60 hz and equal to 200 hz). - Pressure transducer readings are given on the UI - Pressure transducer readings are in the data log - Thermocouple readings are given on the UI - Thermocouple readings are in the data log -
Requirements:	RQ.NLFS.00027, RQ.NLFS.00028, RQ.NLFS.00049, RQ.NLFS.00056, RQ.NLFS.00057, RQ.NLFS.00058
Result:	
Screenshot(s):	
Justification:	

2. HARDWARE

2.1. BOM Component Analysis

Author:	Eli Sells
Executed By:	
Test Date:	
Result:	TODO
Requirements:	RQ.NLFS.00047, RQ.NLFS.00048, RQ.NLFS.00068
Justification/Plan for Remediation (if failed):	

Step 1:

Description:	Verify that COTS products are used.
Steps:	Look at the bill of materials. Verify that: - The LabJack T7 is listed as the primary piece of data acquisition and control hardware - No custom PCBs were designed for use on the test cart.
Requirements:	RQ.NLFS.00047, RQ.NLFS.00048
Result:	
Screenshot(s):	
Justification:	

Step 2:

Description:	Verify that the thermocouple amplifiers are rated to the correct range.
Steps:	Look at the bill of materials.

	Verify that: - The thermocouple amplifiers use external signal conditioning that buffers/amplifies the signal to the LabJack's input range.
Requirements:	RQ.NLFS.00068
Result:	
Screenshot(s):	
Justification:	

2.2. Harnessing Analysis

Author:	Eli Sells
Executed By:	
Test Date:	
Result:	TODO
Requirements:	RQ.NLFS.00051, RQ.NLFS.00062, RQ.NLFS.00064, RQ.NLFS.00065, RQ.NLFS.00070, RQ.NLFS.00071
Justification/Plan for Remediation (if failed):	

Step 1:

Description:	Verify that connectors are labeled.
Steps:	Look at each connector. Verify that: - Each connector to an input/output interface is labeled - Each connector to a power input is labeled - Sensor/actuator connectors terminate at screw terminals, and the terminals from the BOM are rated for 25 A / 300 V. - Cable entries into enclosures incorporate strain-relief cord grips or cable glands.

	- Analog sensors use STP cabling - Other connectors are labeled *Note for test runner: If even a single connector fails, you must fail the test and note it in your justification. We will go back and note the open problem to be fixed in the next release OR fix it and re-run the test.*
Requirements:	RQ.NLFS.00051, RQ.NLFS.00062, RQ.NLFS.00064, RQ.NLFS.00065, RQ.NLFS.00071
Result:	
Screenshot(s):	
Justification:	

Step 2:

Description:	Verify that connectors enclosures are grounded.
Steps:	For each metal piece on the enclosure or that is not used in the actual circuitry, attach a multimeter between the metal and the ground reference. Verify that: - The measured resistance is 10 ohms or less. If there is no metal used in the enclosure, mark this as a “fail” since the enclosure does not meet the specifications.
Requirements:	RQ.NLFS.00070
Result:	
Screenshot(s):	
Justification:	

2.3. Schematic Analysis

Author:	Eli Sells
Executed By:	
Test Date:	
Result:	TODO
Requirements:	RQ.NLFS.00051, RQ.NLFS.00062, RQ.NLFS.00064
Justification/Plan for Remediation (if failed):	

Step 1:

Description:	Solenoid actuators are designed properly. *Note: This procedure should be run by an electrical engineer who understands how to read these schematics.*
Steps:	Open the schematics, and look at the solenoid drivers. Verify that: - The driver circuits include TVS or diode protection across the coil.
Requirements:	RQ.NLFS.00051, RQ.NLFS.00062, RQ.NLFS.00064
Result:	
Screenshot(s):	
Justification:	

2.4. Hardware Error Handling

Author:	Eli Sells
Executed By:	
Test Date:	
Result:	TODO
Requirements:	RQ.NLFS.00052, RQ.NLFS.00053 FC.NLFS.LJ.1, FC.NLFS.LJ.2
Justification/Plan for Remediation (if failed):	

Step 1:

Description:	Killswitch functionality
Steps:	Power up the system and actuate a valve so that the system is not in a safe state. Trigger the killswitch. Verify that: - The system returns the actuated valve to its safe position.
Requirements:	RQ.NLFS.00052, RQ.NLFS.00053, FC.NLFS.LJ.1
Result:	
Screenshot(s):	
Justification:	

Step 2:

Description:	Watchdog timer
Steps:	Power up the system and connect LQD-DAQ. Actuate a valve, then forcefully terminate LQD-DAQ. Verify that: - The system returns the actuated valve to its safe position.

Requirements:	FC.NLFS.LJ.2
Result:	
Screenshot(s):	
Justification:	